

AptLens

Smart Recruitment Matching Platform

FULL TECHNICAL STACK DOCUMENT

Frontend, Backend, Database, Languages, APIs, Services, Libraries and Tooling

Submission Basis

Prepared from the public GitHub main branch after final promotion. Repository:
<https://github.com/mhdmusaddiq/DevSphere-Smart-Recruitment-Matching-Platform>

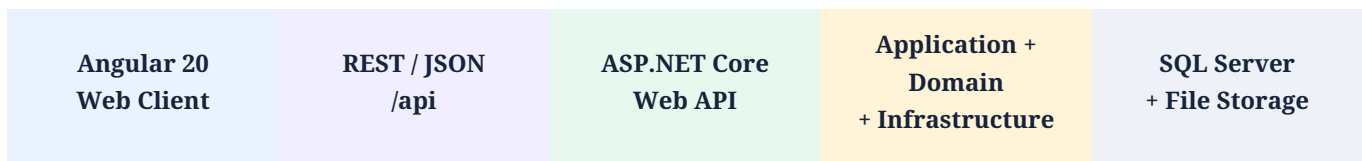
Project	AptLens / DevSphere Smart Recruitment Matching Platform
Repository branch	main
Verified main commit	3a77c235e32af240c38a4b56a89ec63ea089a8e4
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1. Technical Stack at a Glance

AptLens is a full-stack recruitment platform with a role-based Angular web client, an ASP.NET Core REST API, Entity Framework Core persistence, SQL Server storage, JWT-based authentication, document handling, matching workflows, and administrative tooling.

Area	Technology / Detail
.NET version	.NET 8 / target framework net8.0; repository SDK pin 8.0.423
Frontend framework	Angular 20.3.31
Backend framework	ASP.NET Core Web API on .NET 8
Database	Microsoft SQL Server / SQL Server Express with Entity Framework Core
Primary languages	C#, TypeScript, HTML, CSS, JSON
Authentication	ASP.NET Core Identity + JWT Bearer authentication + role authorization
API style	RESTful HTTP API with Swagger / OpenAPI documentation
Document processing	PDF upload validation and parsing using PdfPig
Testing	xUnit on backend; Jasmine + Karma on frontend
Source control	Git + GitHub

System Architecture



The frontend calls the API through a relative /api base URL. During local development, Angular proxies /api to the ASP.NET Core HTTPS endpoint.

2. Frontend Technology

The user interface is implemented as an Angular single-page application with TypeScript, Angular Router, Reactive Forms, Angular HTTP services, standalone components, and custom CSS.

Frontend item	Version	Purpose
Angular Core / Common / Forms / Router	20.3.31	Application framework, routing, forms and UI composition

TypeScript	5.9.3	Primary frontend programming language
RxJS	7.8.x	Observable-based asynchronous and HTTP workflows
Zone.js	0.15.x	Angular runtime integration
Plus Jakarta Sans	5.3.x	Application typography package
Angular CLI	20.3.37	Build, serve and project tooling
Angular Build Tooling	20.3.37	Production/development compilation

Frontend styling and UI approach

- Component-scoped CSS plus global application CSS.
- Responsive layouts for desktop, tablet and mobile.
- Custom shared UI primitives for brand, icons, account menus, matching explanations and role shells.
- No Tailwind CSS or Bootstrap runtime dependency is present in the final package manifest.
- Frontend assets are served from the Angular public directory.

Frontend testing stack

- Jasmine 5.6.x for unit test specifications.
- Karma 6.4.x test runner.
- karma-chrome-launcher 3.2.x for browser execution.
- karma-coverage 2.2.x for coverage integration.

3. Backend Technology

The backend is a layered ASP.NET Core Web API targeting .NET 8. The solution separates API, Application, Domain and Infrastructure responsibilities.

Backend item	Version	Purpose
ASP.NET Core Web API	.NET 8	HTTP API, middleware, controllers, routing and hosting
Microsoft.AspNetCore.Authentication.JwtBearer	8.0.29	JWT bearer authentication
Microsoft.AspNetCore.Identity.EntityFrameworkCore	8.0.29	Identity users, roles and password management
Entity Framework Core SQL Server	8.0.29	SQL Server data access
Entity Framework Core Design	8.0.x	Migrations and design-time tooling
Microsoft.AspNetCore.OpenApi	8.0.29	OpenAPI integration
Swashbuckle.AspNetCore	6.6.2	Swagger UI / API documentation
PdfPig	0.1.16	PDF structure parsing and validation

Backend architecture layers

Area	Technology / Detail
DevSphere.Api	Controllers, middleware, CORS, Swagger, authentication pipeline and service registration.
DevSphere.Application	Application contracts, DTOs and interfaces.
DevSphere.Domain	Core domain types and business model.
DevSphere.Infrastructure	EF Core persistence, Identity, repositories, matching engine, files, email and business services.

4. Database and Persistence

Area	Technology / Detail
Database engine	Microsoft SQL Server; local configuration targets SQL Server Express (SQLEXPRESS)
Database name	DevSphereDb

ORM	Entity Framework Core
Migrations	EF Core migrations stored in DevSphere.Infrastructure/Data/Migrations
Authentication storage	ASP.NET Core Identity tables managed through EF Core
Connection mode	Windows trusted connection with TrustServerCertificate enabled for local development

The backend registers DevSphereDbContext with the SQL Server provider. Repository classes and application services operate on the relational data model through this context.

5. Programming Languages and Data Formats

Language / format	Where used	Role
C#	Backend	API controllers, services, Identity, persistence, matching and tests
TypeScript	Frontend	Angular components, services, route logic and tests
HTML	Frontend	Angular templates and semantic UI structure
CSS	Frontend	Responsive styling, component themes and shared shells
JSON	API/configuration	REST payloads, appsettings, Angular configuration and proxy configuration
SQL (via EF Core)	Database	Relational persistence generated/executed through EF Core

6. APIs and Services Used

Area	Technology / Detail
REST API	ASP.NET Core controllers expose JSON endpoints under /api.
Swagger / OpenAPI	Interactive API documentation is enabled in Development mode.
Angular HTTP client	Frontend services and components call backend endpoints through the shared /api base URL.
JWT authentication service	Bearer tokens protect authenticated routes and are validated on the server.
Identity service	User accounts, password rules, roles and security stamps are managed through ASP.NET Core Identity.
Email verification / recovery	Development uses logged one-time codes; SMTP transport is available for configured non-development environments.
Local file storage	Uploaded PDF files are stored under the configured application file storage root.
PDF validation service	PDF content is structurally validated before accepted storage/use.
Matching service	Backend matching engine and persistence services calculate and store recruitment match results.
Notification / contact services	Role-specific notifications and consent-based contact request workflows are implemented server-side.

Security-related services

- CORS restricted to the Angular local development origins.
- HTTPS redirection in the ASP.NET Core pipeline.
- JWT issuer, audience, lifetime and signing key validation.
- Role-based authorization for Admin, Employer and Candidate/Job Seeker workflows.
- Authentication request rate limiting and verification/recovery challenge limits.
- Session invalidation using Identity security-stamp validation.
- File extension, content type, size and PDF structure checks for uploaded documents.

7. Other Tools, Libraries and Technologies

Tool / library	Area	Use
Git	Source control	Local version control and branch

		workflow
GitHub	Repository / collaboration	Public source repository, pull requests and merge history
dotnet CLI	Backend tooling	Restore, build, test and run .NET solution
dotnet-ef / EF Core migrations	Database tooling	Create and apply schema migrations
npm	Frontend tooling	Dependency installation and script execution
Angular CLI	Frontend tooling	Development server and production builds
Swagger UI	API tooling	Interactive endpoint inspection
xUnit 2.5.3	Backend testing	Automated unit/integration tests
EF Core InMemory 8.0.29	Backend testing	Isolated persistence tests
coverlet.collector 6.0.0	Backend testing	Code-coverage collection support
Jasmine / Karma	Frontend testing	Angular browser-based unit testing

8. Local Development Runtime

Area	Technology / Detail
Frontend URL	http://localhost:4200
Backend HTTPS URL	https://localhost:7097
Backend HTTP URL	http://localhost:5129
Swagger	https://localhost:7097/swagger in Development
Frontend API path	/api
Angular development proxy	/api -> https://localhost:7097
Local file root	App_Data/Files
Configured PDF max size	5 MiB

Note on credentials and secrets

The final backend does not hard-code a production JWT secret or administrator password. JWT signing keys and

optional bootstrap administrator credentials are expected from protected runtime configuration / environment variables.

9. Quality and Testing Technologies

The repository contains both backend and frontend automated tests. The backend test project targets .NET 8 and uses xUnit, EF Core InMemory and coverlet. The Angular project uses Jasmine and Karma with Chrome-based execution.

- Backend tests cover security, files, vacancy lifecycle, matching, authentication and domain workflows.
- Frontend tests cover authentication, route behavior, HTTP interception, vacancy editing and major user-interface logic.
- Production frontend builds use explicit initial-bundle and component-style budgets.
- The final frontend package manifest contains Angular 20 runtime dependencies and separate development-only test/build dependencies.

10. Technical Stack Summary for Submission

Area	Technology / Detail
.NET Version	.NET 8; SDK 8.0.423; projects target net8.0
Frontend Technology / Framework	Angular 20.3.31, TypeScript 5.9.3, RxJS, Angular Router, Reactive Forms, custom CSS
Backend Technology / Framework	ASP.NET Core Web API, C#, Entity Framework Core, ASP.NET Core Identity
Database	Microsoft SQL Server / SQL Server Express; DevSphereDb
Programming Languages	C#, TypeScript, HTML, CSS, JSON
APIs / Services	REST API, Swagger/OpenAPI, JWT auth, Identity, SMTP-capable verification/recovery, local PDF storage, matching and notification services
Other Tools / Libraries	Git, GitHub, npm, Angular CLI, dotnet CLI, EF migrations, PdfPig, Swashbuckle, xUnit, Jasmine, Karma, coverlet

Repository authority

This document was prepared from the actual public main-branch project configuration and package/project manifests, rather than from the older README technology summary.

Source Files Verified

- global.json

- frontend/package.json
- frontend/angular.json
- frontend/proxy.conf.json
- frontend/src/app/core/config/api-base-url.ts
- backend/src/DevSphere.Api/DevSphere.Api.csproj
- backend/src/DevSphere.Infrastructure/DevSphere.Infrastructure.csproj
- backend/src/DevSphere.Application/DevSphere.Application.csproj
- backend/tests/DevSphere.Tests/DevSphere.Tests.csproj
- backend/src/DevSphere.Api/Program.cs
- backend/src/DevSphere.Api/Extensions/ServiceCollectionExtensions.cs
- backend/src/DevSphere.Api/appsettings.json